all other samples have a value of zero. For this reason, the delta function is frequently called the **unit impulse**.

The second term defined in Fig. 6-1 is the **impulse response**. As the name suggests, the impulse response is the signal that exits a system when a delta function (unit impulse) is the input. If two systems are different in any way, they will have different impulse responses. Just as the input and output signals are often called $x[n]$ and $y[n]$, the impulse response is usually given the symbol, $\boldsymbol{h[n]}$. Of course, this can be changed if a more descriptive name is available, for instance, $f[n]$ might be used to identify the impulse response of a *filter*.

Any impulse can be represented as a *shifted* and *scaled* delta function. Consider a signal, $a[n]$, composed of all zeros except sample number 8, which has a value of -3. This is the same as a delta function shifted to the right by 8 samples, and multiplied by -3. In equation form: $a[n] = -3\delta[n-8]$. Make sure you understand this notation, it is used in nearly all DSP equations.

If the input to a system is an impulse, such as $-3\delta[n-8]$, what is the system's output? This is where the properties of homogeneity and shift invariance are used. Scaling and shifting the input results in an identical scaling and shifting of the output. If $\delta[n]$ results in $h[n]$, it follows that $-3\delta[n-8]$ results in $-3h[n-8]$. In words, the output is a version of the impulse response that has been *shifted* and *scaled* by the same amount as the delta function on the input. If you know a system's impulse response, you immediately know how it will react to *any* impulse.

Convolution

Let's summarize this way of understanding how a system changes an input signal into an output signal. First, the input signal can be decomposed into a set of impulses, each of which can be viewed as a scaled and shifted delta function. Second, the output resulting from each impulse is a scaled and shifted version of the impulse response. Third, the overall output signal can be found by adding these scaled and shifted impulse responses. In other words, if we know a system's impulse response, then we can calculate what the output will be for any possible input signal. This means we know *everything* about the system. There is nothing more that can be learned about a linear system's characteristics. (However, in later chapters we will show that this information can be represented in different forms).

The impulse response goes by a different name in some applications. If the system being considered is a *filter*, the impulse response is called the **filter kernel**, the **convolution kernel**, or simply, the **kernel**. In image processing, the impulse response is called the **point spread function**. While these terms are used in slightly different ways, they all mean the same thing, the signal produced by a system when the input is a delta function.